

Voice Quality in Forensic Phonetics: When, How, and Why?

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Forensic phonetics applies phonetic knowledge to solve legal problems, analyzing speech as evidence with a focus on speaker identity. Since voice quality is a powerful carrier of identity, it is particularly valued in tasks such as speaker profiling, the design of voice line-ups, and forensic voice comparison. This keynote introduces the field of forensic phonetics and its main areas of application, with a focus on forensic voice comparison, where recordings of an unknown offender and a suspect are analyzed to determine whether they belong to the same speaker. Forensic practitioners draw on both auditory and acoustic methods, considering not only segmental features but also suprasegmental and prosodic characteristics, including voice quality.

The second part of the talk centers on voice quality, a key forensic parameter that has received increasing attention in recent years. For instance, it is included in the Best Practice Manual for Forensic Speaker Comparison by the European Network of Forensic Science Institutes (ENFSI). Two complementary perspectives are considered regarding voice quality: a narrow definition, focusing on vocal fold activity (typically analyzed through sustained vowels), and a broader one, where voice quality emerges from the interaction of laryngeal and supralaryngeal features. Particular attention is given to one commonly used protocol for the perceptual evaluation of voice quality: the Vocal Profile Analysis (VPA) scheme and its simplified versions, which have proven especially valuable in applied contexts and computer-assisted analyses. Drawing on an international survey of practitioners, the keynote highlights how voice quality is currently conceptualized and used in forensic casework.

The final part briefly addresses the growing challenge of voice deepfakes in forensic phonetics. As synthetic speech becomes more realistic, voice quality must be reconsidered as a cue to identity, calling for closer integration between phonetic expertise and emerging technologies.

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